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Impact of Magnification, Image Type, and Number on Convolutional Neural Network Performance in Differentiating Canine Large Cell Lymphoma From Non-Lymphoma via Lymph Node Cytology

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ABSTRACT

Background: Lymph node (LN) aspirates are often obtained to distinguish large-cell lymphoma (LCL) from non-lymphoma (NL) in dogs with enlarged lymph nodes.

Objective: Images from cytology slides tested the effects of magnification, image type, and number on a convolutional neural network (CNN) differentiating canine LCL from NL.

Methods: Three hundred images of LCL and NL were used to train a CNN and interrogate the effects of image magnification, type, and number on the model's performance. Identified cases were imaged at 200×, 500×, and 1000× magnification in color and gray-scale and then used to train and identify optimal magnification and image type. The impact of the image number per cohort (50, 100, 150, 200, 250, 300) on the top model's performance was then assessed.

Results: The highest performance with color images was achieved at 1000× magnification, with an accuracy of 95.8%, a Receiving Operating Characteristic (ROC) area of 0.997, and an *F*-measure of 0.958. Similarly, the best results with gray images, also at 1000× magnification, showed an accuracy of 96.67%, a ROC area of 0.994, and an *F*-measure of 0.967. Performance improvements were most significant and plateaued as the number of images per class approached 150, with an accuracy of 95%, ROC area of 0.939, and *F*-measure of 0.95.

Conclusion: The analysis across models suggests that color versus greyscale did not significantly impact overall performance to distinguish LCL or NL. Optimal magnification was 1000×. A minimum of 150 images per class is recommended for pilot CNN studies in this 2-class problem.

1 | Introduction

In recent years, the medical field has seen rapid growth in artificial intelligence systems (AIS) for use in clinical and research settings [1]. The recent successes of AIS in solving complex problems within medicine can be attributed to advances in computational power, more cost-efficient data storage, improvements in AIS accuracy, explainability, and interpretability. These advances and the increased accessibility of slide digitalization have allowed for AIS advances in the field of computational pathology [2, 3].

Despite cytology being the first field of pathology in which AIS was implemented, most research to date has focused on histopathology [4]. In anatomic pathology, these AIS have been shown to diagnose, subtype successfully, and grade cancer, count mitotic figures, predict prognosis, and determine genomic instability [5, 6]. In hematopathology, these models successfully perform differential counts, detect poikilocytosis, and detect neoplasia [7]. In cytopathology, one of the earliest uses of AIS was for cervical smears [8]. Since then, AIS have been used to successfully evaluate urine samples and breast, gastrointestinal, and thyroid lesions [8].

Cytology samples can be more challenging to work with than histopathology regarding AIS due to blood obscuring cell morphology, variable cellularity, and thickness (3D layering of cells) of the sample [9, 10]. This variation in thickness can lead to unfocused digital images, making AIS imaging recognition more difficult [9]. Improved whole slide image technology has helped address this problem, allowing for more AI research in cytopathology.

Cytology is a standard diagnostic tool for many diseases in veterinary medicine, including lymphoma [11]. A specific type of AIS, convolutional neural networks (CNN), has been shown to perform very well in classifying images from various diseases [4, 9, 10]. This study aims to design a CNN that can differentiate canine large-cell lymphoma from non-lymphoma using images collected from cytology samples. This study evaluated the effects of magnification, image type (color versus 16-gray bit), and number of images on the accuracy of the CNN, laying the groundwork for future AIS studies in the field of veterinary cytopathology.

2 | Materials and Methods

We conducted a retrospective review at the Virginia-Maryland College of Veterinary Medicine covering the period from 2018 to 2023. The initial pool consisted of 500 canine patients undergoing peripheral lymph node aspiration. The inclusion criteria required no prior history of lymphoma and sample cellularity ranging from moderate to high, with good to excellent preservation. One of five board-certified veterinary clinical pathologists confirmed each diagnosis, categorizing samples into two primary groups: large-cell lymphoma (LCL; 150 cases) and non-lymphoma (NL; 150 cases). Categorization of "large-cell" lymphoma was based on the cell size being greater than the size of a neutrophil. In the NL group, there were 95 cases of reactive lymphoid hyperplasia, 44 cases of reactive lymphoid hyperplasia with lymphadenitis, 6 cases of lymphadenitis, and 5 cases of cytologically unremarkable lymph nodes. Detailed patient information, such as age, sex, and breed, was extracted from medical records.

This was a retrospective study, and all included cases had a preexisting cytologic diagnosis made by a board-certified ACVP Diplomate in pathology. For inclusion in the LCL group, each case was re-reviewed microscopically by both the resident (PI) and the Diplomate author to ensure concordance with the original diagnosis. Additionally, a subset of 20 of the LCL cases was assessed via PCR for Antigen Receptor Rearrangement (PARR) to confirm clonality. Of these cases, 19 were found to be monoclonal and one had no amplification.

2.1 | Image Acquisition and Pre-Processing

A single slide was selected for each of the 300 cases that met the inclusion criteria. All slides had been stained with a Wright-Giemsa stain and stored without a coverslip. Only those slides reported as moderate or high cellularity with good to excellent cellular preservation were included. Each slide was then divided into eight evenly distributed quadrants labeled 1–8 (see Figure 1), and evaluated for cellularity, number of intact cells, staining of cells, and arrangement of cells. In quadrants with moderate or high cellularity, mostly intact cells, adequate staining, and a monolayer arrangement, a single image was taken at 200×, 500× (oil immersion), and 1000× (oil immersion) magnifications, respectively. If a quadrant was deemed adequate for imaging at 500× and 1000× magnification, an image was taken at 200× magnification, even if the field of view had lower cellularity than desired. The number of lymphocytes per region of interest (ROI) was not standardized and varied depending on the cellularity and magnification of each slide. Each ROI was screened to ensure cytologic features consistent with a lymphoid neoplasm or a non-neoplastic lymph node, in alignment with standard diagnostic practice.

Two veterinary students were used to acquire images. Each veterinary student was assigned different slides for quadrant evaluation and image acquisition, so no inter-observer comparison or reconciliation was necessary. Prior to imaging, all students received standardized training in field selection from the PI.

All slides used in this study were imaged without coverslips. The images were taken with a Canon EOS REBEL T2i with the associated EOS Utility 2 (Version 2.14.20.0) software, attached to either a Nikon Eclipse E400 or an Olympus BX41 microscope. All images were 5184 pixels × 3456 pixels with a horizontal and vertical resolution of 72 × 72 dpi, respectively. Table 1 outlines the microscope objectives and ocular eyepieces.

ImageJ (Java 1.8.0_322) was used for all image modifications. The 'Enhance Contrast' function was used with a 'saturated pixel' setting of 0.35% to normalize image intensity when color images were used in the model. Similarly, ImageJ was used to transform the

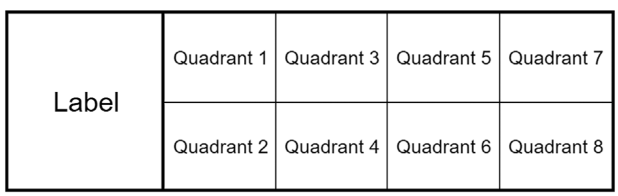

FIGURE 1 | Diagram of a slide divided into 8 even quadrants. A template slide was used to draw on the back of the slides included in the study. Each quadrant was roughly 1.38 cm × 1.25 cm.

images into a 16-bit gray scale, followed by an 'Enhance Contrast,' 'saturated pixel' setting of 0.35%, and image normalization.

The images were used to evaluate 14 CNN models (see Table 2) for the following criteria: (1) ideal magnification and (2) image type (standard red, green, blue color vs. 16-gray bit). The two image types were chosen to evaluate how the dimensionality (3 color channels vs. 2 for gray-scale) affects a CNN. Each model used 600 images (300 large-cell lymphoma images from 150 patients and 300 non-lymphoma images from 150 patients). Example images for large-cell lymphoma and non-lymphoma can be seen in Figures 2 and 3, respectively. Depending on the number of images needed for a model, one to two images from each patient were used (i.e., for model 1A, 300 images at 200× magnification were taken of large-cell lymphoma, and 300 images at 200× magnification were taken of non-lymphoma, but for model 7A, 100 images at 200× magnification, 100× images at 500× magnification, and 100 images at 1000× oil magnification were used so that total images per class was always 300). See Figures 4–6 for a breakdown of each model's case and image inclusion process. When a random number generator was required, either the Excel randomizer function or MATLAB randomizer was used (see Figures 4–6) [12, 13].

Image augmentation through 180° rotation was used to increase dataset variability. This approach is consistent with standard CNN training practices and is not considered a limitation. Rotated images are treated as independent by the algorithm, contributing to model robustness [14].

The models were compared based on their accuracy, weighted Receiving Operating Characteristic (ROC) area, and weighted *F*-measure. In general, an ROC of 0.5 suggests no discrimination (i.e., ability to diagnose patients with and without the disease or condition based on the test), 0.7–0.8 is considered acceptable, 0.8–0.9 is considered excellent, and more than 0.9 is considered outstanding. ROC analysis provides a means to select optimal models and discard suboptimal ones independently from (and before specifying) the cost context or the class distribution.

The *F*-measure is a way of evaluating the predictive performance of a model by using precision and recall. The closer the *F*-measure is to 0, the worse the performance of the model at recall and/or precision. An *F*-measure of 1 would indicate an algorithm with perfect recall and precision. Unlike accuracy, *F*-measure focuses on the false negative and false positive values rather than the total correctly classified incidences.

The color model with the highest accuracy, ROC, and *F*-measure was then used to evaluate the effect of the number of images on the model's performance. Images were added to the two best models in increments of 50 per class (i.e., starting with 50 images per class and increasing to 350 images per class) until a performance plateau was reached.

For each model, the precision, *F*-measure, recall, true positive rate, false positive rate, ROC area, Precision-recall curve area, the Matthews correlation coefficient, correctly classified image percent (accuracy), incorrectly classified image percent, kappa statistic, mean absolute error, root mean squared error, relative absolute error, and root relative squared error were recorded (Table 5). While the models were evaluated based on accuracy, ROC area, and *F*-measure, the other performance assessments from Weka (software) were recorded in Tables 3, 4 and 6 for transparency and completeness.

TABLE 1 | Details of microscope objectives and eyepiece lenses.

Microscope objective		Oil	Manufacturer	Field diameter (mm)
1	20	Not Applicable	Nikon	1.1
	50	Oil	Nikon	0.44
	100	Oil	Nikon	0.22
2	20	Not Applicable	Olympus	1.1
	50	Oil	Olympus	0.44
	100	Oil	Olympus	0.22

[a]Coverslips were not used for any objective. All lenses were 10× and FN 22 mm.

TABLE 2 | Displays the 14 models interrogated to determine the objective and image type that yields the best performance as measured by the *F*-score and ROC area using 300 images per class.

		Model 1	Model 2	Model 3	Model 4	Model 5 Magnification	Model 6	Model 7
	Image type	200×	500×	1000×			200× + 500× 200× + 1000× 500× + 1000× 200× + 500× + 1000×	
A	Red, green, blue	1A	2A	3A	4A	5A	6A	7A
B	16-gray bit	1B	2B	3B	4B	5B	6B	7B

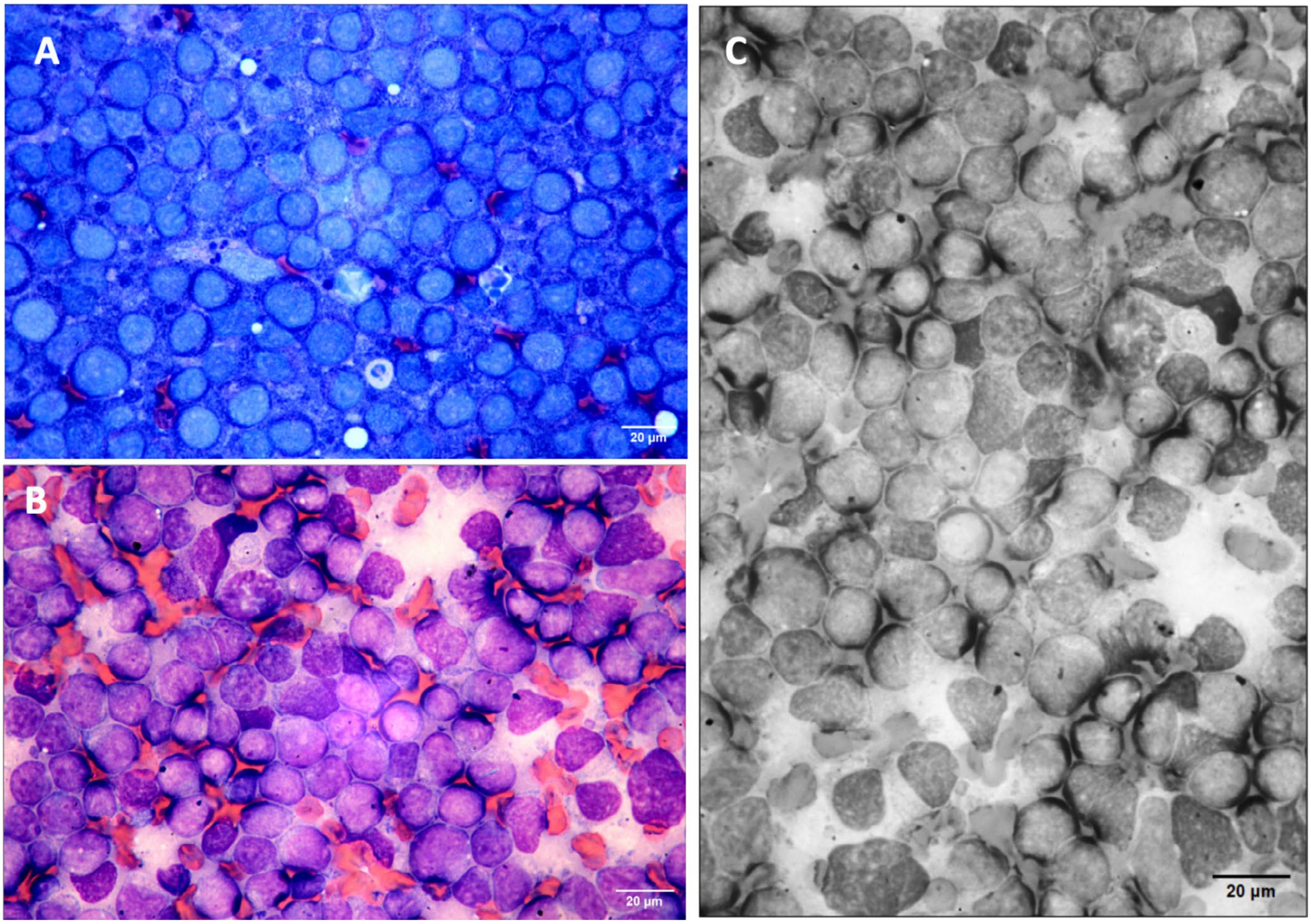

FIGURE 2 | Representative images of large-cell lymphoma used to construct the CNN. (A, B) are color (Wright-Giemsa, 1000× magnification). (C) is the 16-gray bit version of (B), rotated 90° to the right.

2.2 | Convolutional Neural Network Construction (Hardware and Software Requirements)

The WekaDeeplearning4j package for the Weka workbench—Data Mining with Open-Source Machine Learning (version 3.8.6) was used to construct the convolutional neural network. The Weka workbench and ImageJ were used on a Dell Precision 3660, 13th Gen Intel(R) Core (TM) i9-13900K with 32 gigabit 4800 memory, an NVMe PC SN810 NVMe WDC 1024GB hard drive, and an NVIDIA RTX A4500 video card.

The convolutional neural network design and hyperparameter selection were based on the literature [15, 16]. The hyperparameters for all 16 models were fixed. Table 5 shows the hyperparameters used in the WekaDeeplearning4j package. ImageDirectoryLoader was used to upload the images in the required ARFF (Attribute-Related File Format) format.

3 | Results

From the initial cohort of 500 candidates, 300 cases met the inclusion criteria, with 150 cases each in the LCL and NL categories; the selection process ensured a balanced representation of both study groups. Three hundred images per cohort were used to train the CNN, comprised of various magnification combinations (Figures 4–6, 200×, 500×, and 1000×). The distribution of the dogs' age, sex, alteration status, and breeds is listed in Table S1. The median age for the non-lymphoma and large-cell lymphoma groups was 11.5 and 11 years old, respectively. The ages ranged from 3 to 18 years in the non-lymphoma group and 1 to 21 in the large-cell lymphoma group. In both groups, most dogs were male neutered, followed by female spayed canines, with mixed-breed dogs representing the most common breed.

The maximum storage time for slides used for images was 6 years with no coverslip, and no subjective degradation of staining, cellular preservation, or slide quality was observed compared to cases stored for 1 year. The mean, median, standard deviation, and range of quadrants imaged per slide for non-lymphoma cases were 4, 3.9, 1.88, and 1–8, respectively. The mean, median, standard deviation, and range of quadrants imaged per slide for the large-cell lymphoma cases were 4, 3.8, 1.83, and1–8, respectively. A single region of interest (ROI) could be found for 16 large-cell lymphoma and seven non-lymphoma cases, resulting in 1 image/case in these instances. To ensure that all cases were equally represented, these images were augmented by rotating 180° to follow the two images per case rule.

The precision, *F*-measure, recall, true positive rate, false positive rate, ROC area, precision-recall curve area, Matthews

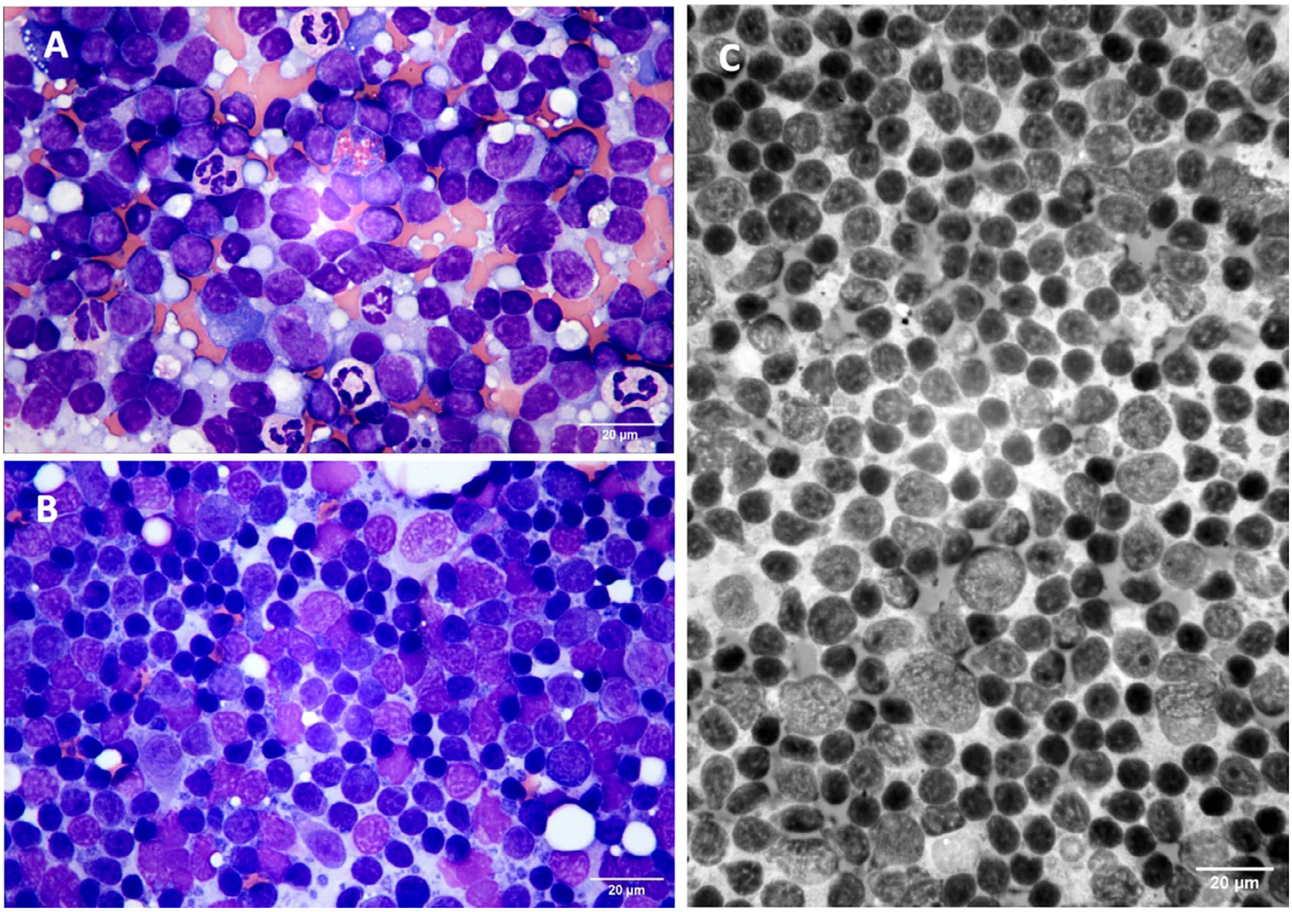

FIGURE 3 | Representative images of non-lymphoma used to construct the CNN. (A, B) are color (Wright-Giemsa, 1000× magnification). (C) is a 16-gray bit example image of non-lymphoma transformed from the original (Wright-Giemsa, 1000× magnification).

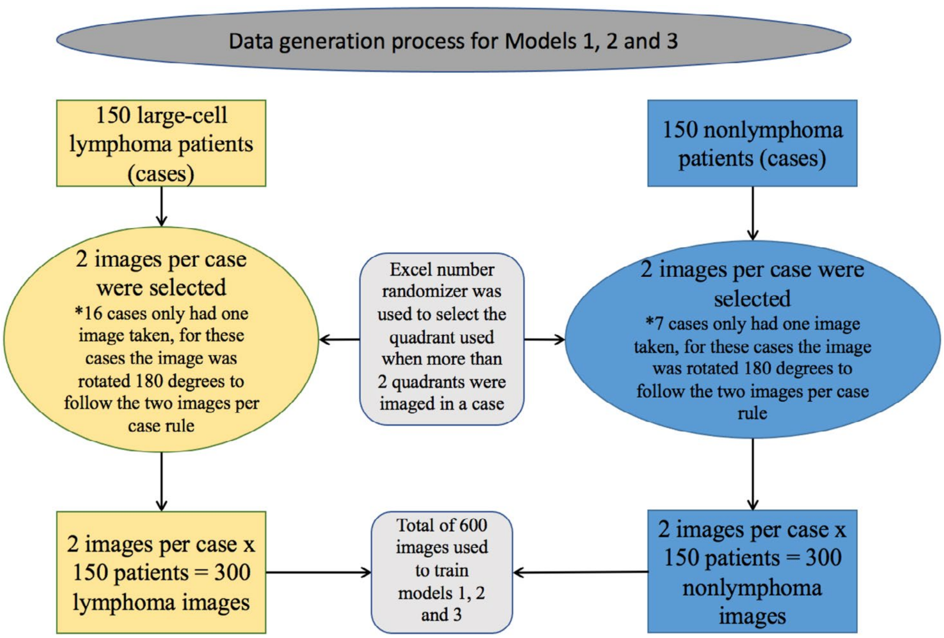

FIGURE 4 | Explanation of case and image inclusion for models 1, 2, and 3.

correlation coefficient, correctly classified image percent (accuracy), incorrectly classified image percent, kappa statistic, mean absolute error, root mean squared error, relative absolute error, and root relative squared error for all 14 models are shown in Tables 3 and 4. Each statistical value was obtained using an 80/20 (training/testing) dataset split of the images using Weka's

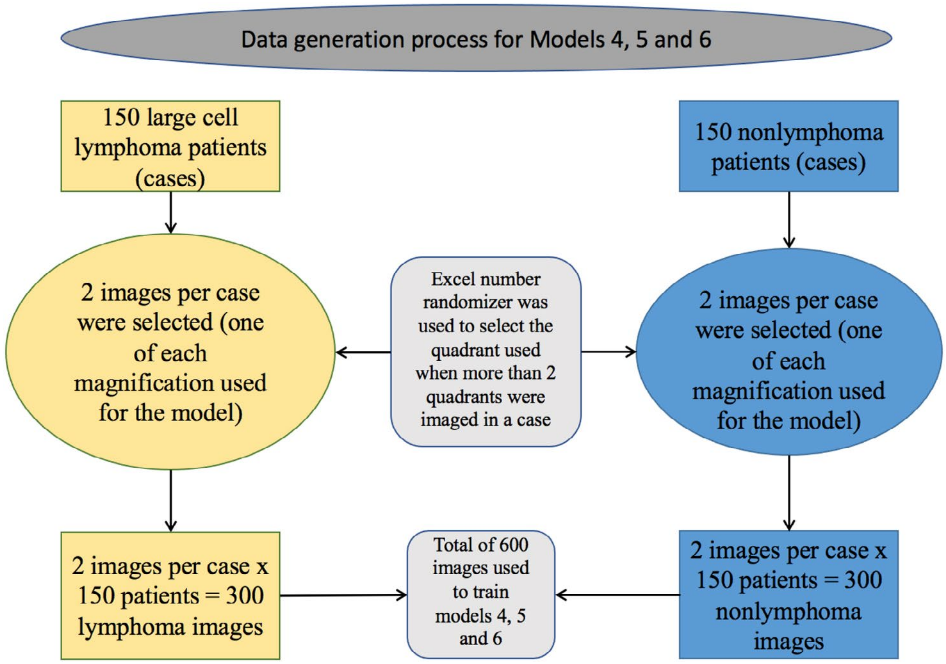

FIGURE 5 | Explanation of case and image inclusion for models 4, 5, and 6.

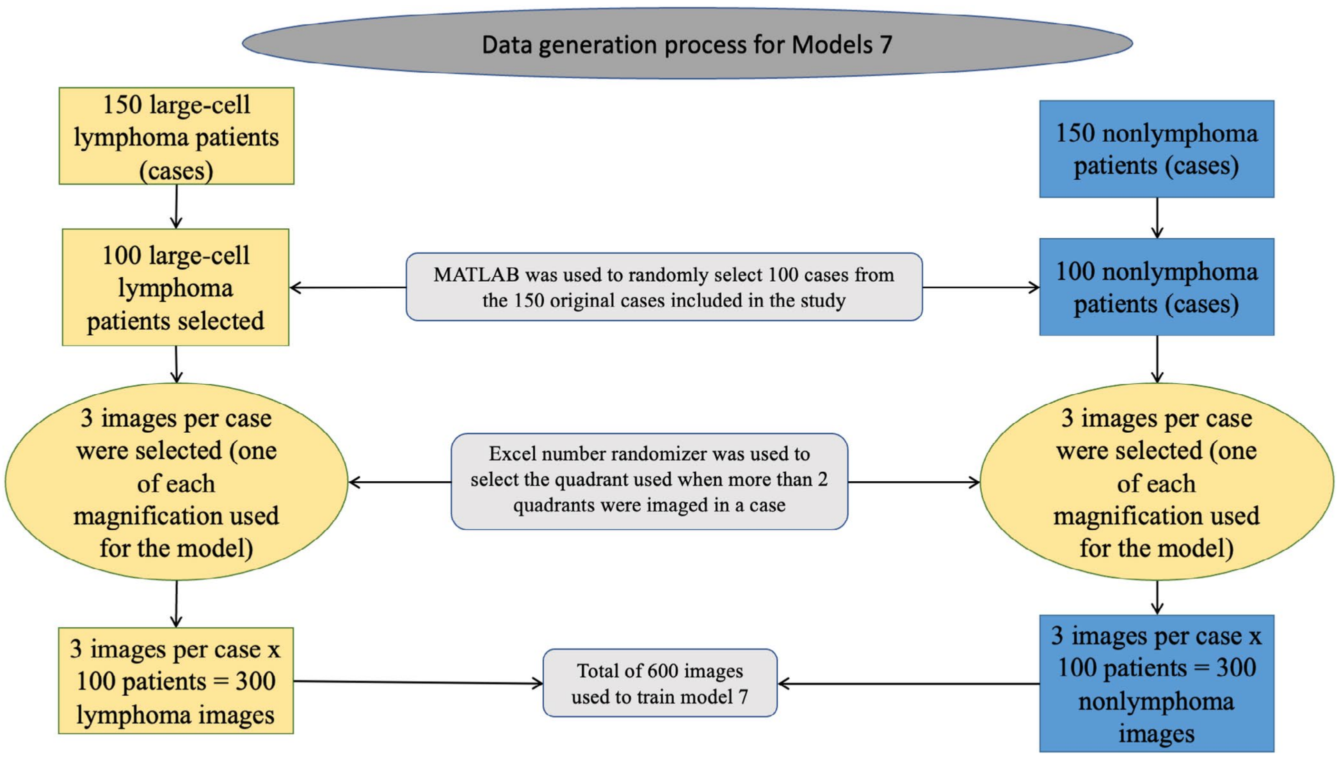

FIGURE 6 | Explanation of case and image inclusion for model 7.

internal randomizer function, so the testing dataset was independent of the training dataset. Table 6 compares the 7 models using 300 color-normalized images each of LCL and NL. The model with the highest performance is model 3A (color normalized; 1000× magnification) with an accuracy of 95.8%, a weighted ROC area of 0.997, and a weighted *F*-score of 0.958. The model that performed second best was model 6A (color normalized images; 500× and 1000× magnifications mixed) with an accuracy of 94.17%, a weighted ROC area of 0.994, and a weighted *F*-measure of 0.942. The model with the lowest performance was model 7A (color normalized images; 200×, 500×, and 1000× magnifications mixed) with an accuracy of 88.33%, a weighted ROC area of 0.959, and an *F*-measure of 0.883.

Table 3 compares the 7 models using 300 16-bit gray normalized images each of LCL and NL. The model with the highest

TABLE 3 | Effects of magnification with 300 16-bit gray images per class on the performance of a CNN to detect large-cell lymphoma versus non-lymphoma using cytology lymph node image. Model 3B (1000× magnification with oil immersion, green) had the best performance based on accuracy, ROC area, and *F*-measure. Model 6B (500× and 1000× magnifications, yellow) had the second-highest performance. In contrast, model 7B (200×, 500×, and 1000× magnifications combined, red) had the lowest performance.

		Model 1B	Model 2B	Model 3B	Model 4B	Model 5B	Model 6B	Model 7B
		Magnification						
16-Gray bit images		**200×**	**500×**	**1000×**	**200× and 500×**	**200× and 1000×**	**500× and 1000×**	**200×, 500× and 1000×**
Precision	Weighted average	0.894	0.917	0.967	0.912	0.918	0.926	0.861
	Lymphoma	0.87	0.909	0.955	0.981	0.897	0.91	0.958
	Nonlymphoma	0.922	0.926	0.981	0.833	0.942	0.943	0.75
F-measure	Weighted average	0.891	0.917	**0.967**	0.9	0.916	**0.925**	**0.832**
	Lymphoma	0.902	0.923	**0.969**	0.898	0.924	**0.931**	**0.821**
	Nonlymphoma	0.879	0.909	**0.964**	0.902	0.907	**0.917**	**0.844**
Recall	Weighted average	0.892	0.917	0.967	0.9	0.917	0.925	0.833
	Lymphoma	0.938	0.938	0.984	0.828	0.953	0.953	0.719
	Nonlymphoma	0.839	0.893	0.946	0.982	0.875	0.893	0.964
True positive rate	Weighted average	0.892	0.917	0.967	0.9	0.917	0.925	0.833
	Lymphoma	0.938	0.938	0.984	0.828	0.953	0.953	0.719
	Nonlymphoma	0.839	0.893	0.946	0.982	0.875	0.893	0.964
False positive rate	Weighted average	0.115	0.086	0.036	0.09	0.089	0.079	0.15
	Lymphoma	0.161	0.107	0.054	0.018	0.125	0.107	0.036
	Nonlymphoma	0.063	0.063	0.016	0.172	0.047	0.047	0.281
ROC area	Weighted average	0.946	0.98	**0.994**	0.954	0.991	**0.969**	**0.919**
	Lymphoma	0.946	0.98	**0.994**	0.954	0.991	**0.969**	**0.919**
	Nonlymphoma	0.946	0.98	**0.994**	0.954	0.991	**0.969**	**0.919**
PRC area (precision-recall curve)	Weighted average	0.95	0.982	0.994	0.935	0.991	0.971	0.895
	Lymphoma	0.946	0.984	0.995	0.971	0.993	0.975	0.945
	Nonlymphoma	0.954	0.979	0.992	0.894	0.989	0.966	0.838

(Continues)

TABLE 3 | (Continued)

16-Gray bit images		Model 1B	Model 2B	Model 3B	Model 4B	Model 5B	Model 6B	Model 7B
		Magnification						
		200×	500×	1000×	200× and 500×	200× and 1000×	500× and 1000×	200×, 500× and 1000×
Matthews correlation coefficient	Weighted average	0.784	0.833	0.933	0.813	0.834	0.85	0.696
	Lymphoma	0.784	0.833	0.933	0.813	0.834	0.85	0.696
	Nonlymphoma	0.784	0.833	0.933	0.813	0.834	0.85	0.696
Correctly classified (percent)		89.17	91.67	**96.67**	90	91.67	**92.5**	**83.33**
Incorrectly classified (percent)		10.83	8.33	3.33	10	8.33	7.5	16.67
Kappa statistic		0.78	0.83	0.93	0.8	0.83	0.85	0.67
Mean absolute error		0.1	0.075	0.042	0.12	0.091	0.092	0.17
Root mean squared error		0.3	0.26	0.18	0.3	0.25	0.26	0.37
Relative absolute error		20.96	15	8.42	21.57	18.21	18.34	34.12
Root relative squared error		59.37	51.01	35.67	58.44	49.13	52.8	73.95

performance is model 3B (16-bit gray normalized; 1000× magnification) with an accuracy of 96.67%, weighted ROC area of 0.994, and weighted *F*-measure of 0.967. The model with the second highest performance was model 6B (16-bit gray normalized; 500× and 1000× magnifications) with an accuracy of 92.5%, a weighted ROC area of 0.969, and a weighted *F*-measure of 0.925. The model with the lowest performance was model 7B (16-bit gray normalized; 200×, 500×, and 1000× magnifications) with an accuracy of 83.33%, a weighted ROC area of 0.919, and a weighted *F*-measure of 0.832.

The best color model (3A, 1000× magnification) was used to evaluate the effects of image number on the accuracy, ROC area, and *F*-measure. A range of image numbers from 50 images per class to 350 images per class was interrogated until a plateau in performance was reached. The precision, *F*-measure, recall, true positive rate, false positive rate, ROC area, precision-recall curve area, Matthews correlation coefficient, correctly classified image percent (accuracy), incorrectly classified image percent, kappa statistic, mean absolute error, root mean squared error, relative absolute error, and root relative squared error for all 7 models are displayed in Table 4. A confusion matrix generated for the model built with 150 images per class is provided in Table S2. Each statistical value was obtained using an 80/20 (training/testing) dataset split into a weighted average and a class average for large-cell lymphoma and non-lymphoma, respectively.

4 | Discussion

This feasibility study evaluates the effects of magnification and image type on the accuracy, ROC area, and *F*-measure. Based on this study, when using CNNs for cytopathology, using 300 color or gray-scale images per class, the 1000× magnification with oil immersion and no coverslip yielded the highest performance, distinguishing NL from LCL with an accuracy of 95.8% and 96.7% respectively. Additionally, the ROC was nearly 100 for both color and gray-scale images at 1000× magnification, indicating high agreement between the algorithm's prediction and the human (manual) method. One advantage of cytopathology is the cellular detail that can be achieved at higher magnifications, so, unsurprisingly, CNN performed best at higher magnifications. Based on this study, when using CNNs for cytopathology, the higher magnification with oil immersion yields the highest performance.

The model performance decreased when all three magnifications were combined (models 7A and 7B). One potential cause for this is that there were fewer images from each magnification for the CNN to learn from. It is hypothesized that if a greater number of images from each magnification were used in the models 7A and 7B CNNs, the performance would have improved.

The accuracy, ROC area, and *F*-measure were highly impacted by the number of images used to train and test the model. Our findings suggest that the minimum number of images recommended for cytopathology is 150 per class to achieve a minimal accuracy of 95%, ROC area of 0.94, and *F*-measure of 0.95. There is a noticeable increase in accuracy, weighted ROC area, and weighted *F*-measure when comparing the results of 100 images

TABLE 4 | Effects of image number on the performance of a CNN to detect large-cell lymphoma versus non-lymphoma using cytology lymph node color images collected at 1000× magnification. The largest performance improvement and then plateau in the *F*-measure and accuracy occurred at 150 images per class (purple). The largest increase in ROC area occurred at 100 images per class (purple).

		1000× magnification, color images						
		Number of images per class						
		50	100	150	200	250	300	350
Precision	Weighted average	0.702	0.814	0.955	0.963	0.93	0.959	0.972
	Lymphoma	0.692	0.739	0.909	0.974	0.915	0.94	0.986
	Nonlymphoma	0.714	0.882	1	0.951	0.943	0.98	0.956
F-measure	Weighted average	0.694	0.799	**0.95**	0.962	0.93	0.958	0.971
	Lymphoma	0.75	0.81	**0.952**	0.962	0.925	0.962	0.973
	Nonlymphoma	0.625	0.789	**0.947**	0.963	0.935	0.954	0.97
Recall	Weighted average	0.7	0.8	0.95	0.963	0.93	0.958	0.971
	Lymphoma	0.818	0.895	1	0.95	0.935	0.984	0.959
	Nonlymphoma	0.556	0.714	0.9	0.975	0.926	0.929	0.985
True positive rate	Weighted Average	0.7	0.8	0.95	0.963	0.93	0.958	0.971
	Lymphoma	0.818	0.895	1	0.95	0.935	0.984	0.959
	Nonlymphoma	0.556	0.714	0.9	0.975	0.926	0.929	0.985
False positive rate	Weighted average	0.326	0.191	0.05	0.038	0.069	0.045	0.027
	Lymphoma	0.444	0.286	0.1	0.025	0.074	0.071	0.015
	Nonlymphoma	0.182	0.105	0	0.05	0.065	0.016	0.041
ROC area	Weighted average	0.768	**0.92**	0.939	0.995	0.986	0.997	0.991
	Lymphoma	0.768	**0.92**	0.938	0.994	0.986	0.997	0.991
	Nonlymphoma	0.768	**0.92**	0.939	0.994	0.986	0.997	0.991
PRC area (precision-recall curve)	Weighted average	0.792	0.921	0.909	0.995	0.987	0.998	0.992
	Lymphoma	0.85	0.895	0.85	0.995	0.984	0.998	0.993
	Nonlymphoma	0.721	0.945	0.968	0.995	0.989	0.997	0.99
Matthews correlation coefficient	Weighted average	0.39	0.615	0.905	0.925	0.859	0.917	0.943
	Lymphoma	0.39	0.615	0.905	0.925	0.859	0.917	0.943
	Nonlymphoma	0.39	0.615	0.905	0.925	0.859	0.917	0.943
Correctly classified (percent)		70	80	**95**	96.25	93	95.8	97.143
Incorrectly classified (percent)		30	20	5	3.75	7	4.2	2.857
Kappa statistic		0.381	0.603	0.9	0.925	0.859	0.916	0.943
Mean absolute error		0.339	0.187	0.058	0.049	0.076	0.0384	0.035
Root mean squared error		0.491	0.397	0.204	0.181	0.241	0.1689	0.166
Relative absolute error		67.643	37.436	11.62	9.775	15.08	7.67	7.084
Root relative squared error		97.93	79.35	40.818	36.272	48.094	33.73	33.196

per class (200 total) with 150 images per class (300 total). While the model performance does continue to increase up to 350 images per class (700 total), the increase was minimal (*F*-measure change was minimal with a delta no greater than 0.02). This finding is similar to the literature, which reports an improvement in model performance of up to 150 images, with slower improvement noted between 150 and 500 images when using CNNs for image classification tasks [17]. For a pilot study or as proof of concept, 150 images per class seem acceptable. However, for CNNs deployed for clinical use in the medical field, more images will be needed to represent the diversity of patients the model will observe.

TABLE 5 | Fixed hyperparameter settings for all models based on the Comparing deep feature extraction strategies for diabetic retinopathy stage classification from fundus images and Deep convolutional neural network Inception-v3 model for differential diagnosing of lymph node in cytological images: A pilot study [15, 16].

Hyperparameter	Setting
Zoo model	Inception-v3
Number of epochs	20
Early stopping	None
Batch size	8
Pretrained model	Imagenet
Instance iterator	Image instance iterator
	Size of minibatch: 20
	Desired width: 5184
	Desired height: 3456
	Desired number of channels: 1
Neural network configuration	Optimization algorithm: stochastic gradient descent
	Updater: Adam; learning rate: 0.01
	biasUpdater: Stochastic gradient descent; learning rate: 0.01
	L2 regularization: 0.0002
	Weight Initializing method: Xavier
	Dropout: 0.2
Attribute normalization	Normalized training data

There are several limitations to this study. First, due to natural variability in slide cellularity, some cases contributed only one region of interest. In cytopathology, interpretations are based on regions with sufficient cellular detail, not the total number of fields per slide. Similarly, the CNN was trained on high-quality, diagnostically relevant images regardless of the number contributed per case. As such, this does not limit the model's validity, as learning in CNNs is image-based rather than case-based.

For this study, if a cellular ROI at 500× or 1000× magnification was identified, then a 200× magnification image was also taken. As a result, there was a greater chance of capturing low cellularity areas, impacting the model's performance using the 200× magnification images. Given this difficulty, the higher magnification images are likely more ideal for cytopathology. Additionally, the decision to use images captured from a camera instead of whole slide imaging (WSI) was made due to data storage requirements needing long-term (i.e., as more cases are imaged/added to improve this work to make the model robust enough for in-clinic use), the WSI requirements would be substantial and beyond the scope of this feasibility study. As WSI becomes more accessible, ROIs may be captured from WSI and used as an alternative to or in conjunction with a microscope camera.

Additionally, the ROC area thresholding used in this study follows common conventions [18, 19]. We acknowledge that more conservative thresholds, such as those described by Çorbacıoğlu and Aksel, are appropriate when assessing medical diagnostic tools for clinical use due to the high-risk nature of veterinary medicine (i.e., potential for euthanasia) [20]. However, as this study represents early model prototyping, our performance metrics are intended solely for internal comparison and optimization, not for clinical inference.

Another limitation of the study is that the large-cell lymphoma cases were not confirmed with histopathology and immunohistochemistry, and the immunophenotype was unknown. The decision to include cases that were not confirmed with these additional diagnostics was made because the standard diagnosis of large-cell lymphoma can be made cytologically with high sensitivity and specificity [11]. Lastly, this study's AIS only examined two cohorts of enlarged lymph nodes in dogs (large-cell lymphoma and non-lymphoma). It is unclear if the image type and number required would be impacted if more disease classifications were included or if the findings are transferable to other species.

This study represents the model development or technical validation phase of development for a CNN to detect large-cell lymphoma from non-lymphoma using cytology images. Additionally, it lays the groundwork for understanding some of the unique requirements for implementing CNNs using cytopathology. Further studies are required to assess the effects on performance metrics of resolution, size of the images, number of cells evaluated, and type of lymphoma and non-lymphoma (lymphadenitis, lymphoid hyperplasia, etc.). Also, these results do not necessarily represent diagnostic performance in a clinical

TABLE 6 | Effects of magnification with 300 color images per class on the performance of a CNN to detect large-cell lymphoma versus non-lymphoma using cytology lymph node image. Model 3A (1000× magnification with oil immersion, green) had the best performance based on accuracy, ROC area, and *F*-measure. Model 6A (500× and 1000× magnifications, yellow) had the second-highest performance. In contrast, model 7A (200×, 500×, and 1000× magnifications combined, red) had the lowest performance.

		Model 1A	Model 2A	Model 3A	Model 4A	Model 5A	Model 6A	Model 7A
		Magnification						
Color images		200×	500×	1000×	200× and 500×	200× and 1000×	500× and 1000×	200×, 500× and 1000×
Precision	Weighted average	0.915	0.924	0.959	0.905	0.935	0.942	0.884
	Lymphoma	0.965	0.875	0.94	0.948	1	0.952	0.903
	Nonlymphoma	0.857	0.979	0.98	0.855	0.862	0.93	0.862
F-measure	Weighted average	0.908	0.916	**0.958**	0.9	0.925	**0.942**	**0.883**
	Lymphoma	0.909	0.926	**0.962**	0.902	0.924	**0.945**	**0.889**
	Nonlymphoma	0.908	0.904	**0.954**	0.898	0.926	**0.938**	**0.877**
Recall	Weighted average	0.908	0.917	0.958	0.9	0.925	0.942	0.883
	Lymphoma	0.859	0.984	0.984	0.859	0.859	0.938	0.875
	Nonlymphoma	0.964	0.839	0.929	0.946	1	0.946	0.893
True positive rate	Weighted average	0.908	0.917	0.958	0.9	0.925	0.942	0.883
	Lymphoma	0.859	0.984	0.984	0.859	0.859	0.938	0.875
	Nonlymphoma	0.964	0.839	0.929	0.946	1	0.946	0.893
False positive rate	Weighted average	0.085	0.093	0.045	0.094	0.066	0.058	0.115
	Lymphoma	0.036	0.161	0.071	0.054	0	0.054	0.107
	Nonlymphoma	0.141	0.016	0.016	0.141	0.141	0.063	0.125
ROC area	Weighted average	0.97	0.996	**0.997**	0.971	0.984	**0.994**	**0.959**
	Lymphoma	0.97	0.996	**0.997**	0.971	0.984	**0.994**	**0.959**
	Nonlymphoma	0.97	0.996	**0.997**	0.971	0.984	**0.994**	**0.959**
PRC area (precision-recall curve)	Weighted average	0.972	0.996	0.998	0.972	0.985	0.994	0.961
	Lymphoma	0.977	0.996	0.998	0.975	0.987	0.995	0.958
	Nonlymphoma	0.967	0.995	0.997	0.97	0.983	0.994	0.965

(Continues)

TABLE 6 | (Continued)

Color images		**Model 1A**	**Model 2A**	**Model 3A**	**Model 4A**	**Model 5A**	**Model 6A**	**Model 7A**
		Magnification						
		200×	**500×**	**1000×**	**200× and 500×**	**200× and 1000×**	**500× and 1000×**	**200×, 500× and 1000×**
Matthews correlation coefficient	Weighted average	0.823	0.839	0.917	0.804	0.86	0.883	0.767
	Lymphoma	0.823	0.839	0.917	0.804	0.86	0.883	0.767
	Nonlymphoma	0.823	0.839	0.917	0.804	0.86	0.883	0.767
Correctly classified (percent)		90.83	91.67	**95.8**	90	92.5	**94.17**	**88.33**
Incorrectly classified (percent)		9.17	8.33	4.2	10	7.5	5.83	11.67
Kappa statistic		0.82	0.83	0.916	0.8	0.8508	0.88	0.77
Mean absolute error		0.097	0.08	0.0384	0.11	0.0864	0.056	0.12
Root mean squared error		0.29	0.25	0.1689	0.3	0.25	0.2	0.3
Relative absolute error		19.27	15.01	7.67	22.1	17.27	11.16	23.9
Root relative squared error		57.85	49.47	33.73	60.7	50.32	40.19	59.85

setting. As the literature suggests, additional validation steps are required for implementation and clinical use [21, 22].

The next stage of development for an AI intended for medical use is external validation (in silico testing). External validation is performed on a new dataset collected at a different time or from a different laboratory than the training/testing dataset used to build and evaluate the model. This phase of development includes evaluating the model for bias (systemic error), uncertainty (random error), repeatability, run-time monitoring (safety features that flag an output or image with an error message, indicating a high probability the model output is/will be unreliable in that instance), and stress testing (evaluate the system's ability to generalize to unseen data). Additionally, the model should be calibrated. This stage of development will provide a greater understanding of the model's limitations (i.e., how it performs when images are collected using a different stain or from a different laboratory).

In conclusion, for a pilot study designed for a 2-class problem using CNNs for cytopathology with an image size of 5184 pixels × 3456, we recommend the Inception-v3 model with a minimum of 150 images per class and 1000× magnification (oil immersion). This permitted distinction of large-cell lymphoma or non-lymphoma with an accuracy of 96%. The image type or dimensionality (three-channel color; two-channel gray-scale) does not affect the model's performance.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** vcp70056-sup-0001-TableS1.xlsx. **Table S2:** vcp70056-sup-0002-TableS2.docx.